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Modeling and Discrete Simulation

Subject 4

Mathematical and Statistical Models in Simulation

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Slides are derived from the course textbook "Banks, Carson, Nelson, Nicol. Discrete-Event System Simulation, 5th Edition, Prentice Hall, 2010" and from textbook's course slides.

Purpose & Overview

- For the modeling of real-world phenomena, in most of the situations events and actions of the system entities are unpredictable.
- The world the model-builder sees is probabilistic rather than deterministic.
  - Some statistical model might well describe the variations.
- An appropriate model can be developed by sampling the phenomenon of interest:
  - Select a known distribution through educated guesses
  - Make estimate of the parameter(s)
  - Test for goodness of fit
- In this lecture:
  - Review several important probability distributions
  - Present some typical application of these models

PROBABILITY REVIEW

Review of Terminology and Concepts

- In this section of the lecture, we will review the following concepts:
  - Discrete random variables
  - Continuous random variables
  - Cumulative distribution function
  - Expectation
  - The mode

Discrete Random Variables

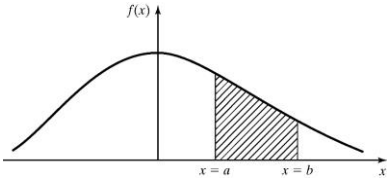
- X is a discrete random variable if the number of possible values of X is finite, or countably infinite.
- Example: Consider jobs arriving at a job shop.
  - Let X be the number of jobs arriving each week at a job shop.
  - R_x = possible values of X (range space of X) = $\{0,1,2,\ldots\}$
  - $p(x_i) = P(X = x_i)$ = probability the random variable is x_i
  - $p(x_i)$, $i = 1,2, \ldots$ must satisfy:
    1. $p(x_i) \geq 0$, for all i
    2. $\sum_{i=1}^{\infty} p(x_i) = 1$
  - The collection of pairs $[x_i, p(x_i)]$, $i = 1,2,\ldots$, is called the probability distribution of X, and $p(x_i)$ is called the ***probability mass function (pmf)*** of X.

Continuous Random Variables

- X is a continuous random variable if its range space R_x is an interval or a collection of intervals.
- The probability that X lies in the interval $[a,b]$ is given by:

$$P(a \le X \le b) = \int_a^b f(x)dx$$

- $f(x)$, denoted as the ***pdf*** of X, satisfies:
  1. $f(x) \ge 0$, for all x in R_X
  2. $\int_{R_X} f(x)dx = 1$
  3. $f(x) = 0$, if x is not in R_X

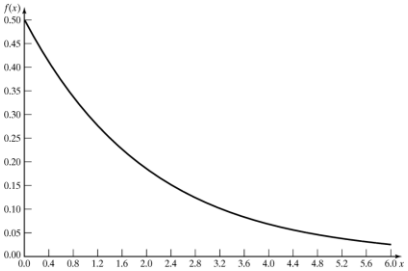


- Properties
  1. $P(X = x_0) = 0$, because $\int_{x_0}^{x_0} f(x)dx = 0$
  2. $P(a \le X \le b) = P(a \prec X \le b) = P(a \le X \prec b) = P(a \prec X \prec b)$

Continuous Random Variables

- **Example:** Life of an inspection device is given by X, a continuous random variable with ***pdf***:

$$f(x) = \begin{cases} \frac{1}{2}e^{-x/2}, & x \ge 0 \\ 0, & \text{otherwise} \end{cases}$$



  - X has an exponential distribution with mean 2 years
  - Probability that the device's life is between 2 and 3 years is:

$$P(2 \le x \le 3) = \frac{1}{2}\int_2^3 e^{-x/2}dx = 0.145$$

Cumulative Distribution Function

- *Cumulative Distribution Function* (***cdf***) is denoted by *F(x)*, measures the probability that the random variable *X* assumes a value less than or equal to *x*, where *F(x) = P(X <= x)*
  - If *X* is discrete, then $F(x) = \sum_{\substack{\text{all} \\ x_i \le x}} p(x_i)$
  - If *X* is continuous, then $F(x) = \int_{-\infty}^{x} f(t)dt$
- Properties
  1. *F* is nondecreasing function. If $a \prec b$, then $F(a) \le F(b)$
  2. $\lim_{x \to \infty} F(x) = 1$
  3. $\lim_{x \to -\infty} F(x) = 0$
- All probability question about *X* can be answered in terms of the ***cdf***, e.g.:

$$P(a \prec X \le b) = F(b) - F(a), \text{ for all } a \prec b$$

Cumulative Distribution Function

- Example: An inspection device has ***cdf***:

$$F(x) = \frac{1}{2}\int_0^x e^{-t/2}dt = 1 - e^{-x/2}$$

  - The probability that the device lasts for less than 2 years:

$$P(0 \le X \le 2) = F(2) - F(0) = F(2) = 1 - e^{-1} = 0.632$$

  - The probability that it lasts between 2 and 3 years:

$$P(2 \le X \le 3) = F(3) - F(2) = (1 - e^{-(3/2)}) - (1 - e^{-1}) = 0.145$$

Expected Value

- The expected value of X is denoted by $E(X)$
  - If X is discrete
    $$E(x) = \sum_{\text{all } i} x_i\, p(x_i)$$
  - If X is continuous
    $$E(x) = \int_{-\infty}^{\infty} x f(x) dx$$
  - a.k.a the *mean*, $\boldsymbol{m}$, or the 1st moment of X, *(also denoted as $\bar{x}$)*
  - A measure of the central tendency

- The variance of X is denoted by $V(X)$ or $var(X)$ or σ^2
  - Definition: $V(X) = E[(X - E[X])^2]$
  - Also, $V(X) = E(X^2) - [E(X)]^2$
  - A measure of the spread or variation of the possible values of X around the mean

- The standard deviation of X is denoted by σ
  - Definition: square root of $V(X)$

Expected Value

- Example: The mean of life of the previous inspection device is:

$$E(X) = \frac{1}{2}\int_0^{\infty} x e^{-x/2} dx = -x e^{-x/2}\Big|_0^{\infty} + \int_0^{\infty} e^{-x/2} dx = 2$$

- To compute variance of X, we first compute $E(X^2)$:

$$E(X^2) = \frac{1}{2}\int_0^{\infty} x^2 e^{-x/2} dx = -x^2 e^{-x/2}\Big|_0^{\infty} + \int_0^{\infty} e^{-x/2} dx = 8$$

- Hence, the variance and standard deviation of the device's life are:

$$V(X) = 8 - 2^2 = 4$$
$$\sigma = \sqrt{V(X)} = 2$$

The mode

- In the discrete statistical models (or discrete probability distributions)
  - The mode is the value of the random variable that occurs most frequently.
- In the continuous statistical models (or continuous probability distributions)
  - The mode is the value at which the ***pdf*** is maximized.
- The mode might not be unique.
  - E.g. bimodal

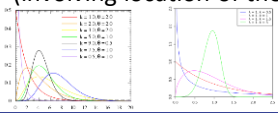
USEFUL MODELS

Useful Statistical Models

- In this section of the lecture, statistical models appropriate to some application areas are presented. The areas include:
  - Queuing systems
  - Inventory and supply-chain systems
  - Reliability and maintainability
- Statistical models are also useful in the case of
  - limited data

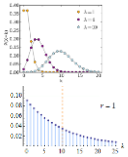
Queuing Systems

- In a queuing system, interarrival and service-time patterns can be probabilistic
- Sample statistical models for interarrival or service time distribution:
  - Exponential distribution: if service times are completely random
  - Normal distribution: fairly constant but with some random variability (either positive or negative)
  - Truncated normal distribution: similar to normal distribution but with restricted value.
  - Gamma and Weibull distribution: more general than exponential (involving location of the modes of *pdf*'s and the shapes of tails.)



Inventory and supply chain

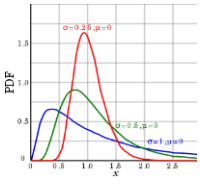
- In realistic inventory and supply-chain systems, there are at least three random variables:
  - The number of units demanded per order or per time period
  - The time between demands
  - The lead time (time between placing of an order and the receipt of that order)
- Sample statistical models for lead time distribution:
  - Gamma
- Sample statistical models for demand distribution:



  - Poisson: simple and extensively tabulated.
  - Negative binomial distribution: longer tail than Poisson (more large demands).
  - Geometric: special case of negative binomial given at least one demand has occurred.

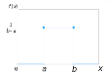
Reliability and maintainability

- Time to failure (TTF)
  - Exponential: failures are random
  - Gamma: for standby redundancy where each component has an exponential TTF
  - Weibull: failure is due to the most serious of a large number of defects in a system of components
  - Normal: failures are due to wear
  - Lognormal: failures only for some type of components



Other areas

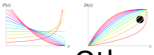
- For cases with limited data, some useful distributions are:
  - Uniform,



    - When an interarrival or service time is known to be random, but no information is immediately available about the distribution.
  - Triangular



    - When there are assumptions about the minimum, maximum, and the modal values.
  - Beta



    - Many distributions which can be shifted to any desired interval.
- Other distribution:
  - Bernoulli,
  - Binomial,
  - Hyperexponential.

DISCRETE DISTRIBUTIONS

Discrete Distributions

- Discrete random variables are used to describe random phenomena in which only integer values can occur.
- In this section of the lecture, we will learn about:
  - Bernoulli trials and Bernoulli distribution
  - Binomial distribution
  - Geometric and negative binomial distribution
  - Poisson distribution

Bernoulli Trials and Bernoulli Distribution

- Bernoulli Trials:
  - Consider an experiment consisting of n trials, each can be a success or a failure.
    - Let $X_j = 1$ if the jth experiment is a success
    - and $X_j = 0$ if the jth experiment is a failure
  - The Bernoulli distribution (one trial):

$$p_j(x_j) = p(x_j) = \begin{cases} p, & x_j = 1,\ j = 1,2,\ldots,n \\ 1-p = q, & x_j = 0, j = 1,2,\ldots,n \\ 0, & \text{otherwise} \end{cases}$$

  - where $E(X_j) = p$ and $V(X_j) = p(1-p) = pq$
- Bernoulli process:
  - The n Bernoulli trials where trials are independent:

$$p(x_1,x_2,\ldots,x_n) = p_1(x_1)p_2(x_2)\ \ldots\ p_n(x_n)$$

Binomial Distribution

- The number of successes in *n* Bernoulli trials, *X*, has a binomial distribution.

$$p(x) = \begin{cases} \binom{n}{x} p^x q^{n-x}, & x = 0,1,2,\ldots,n \\ 0, & \text{otherwise} \end{cases}$$

The number of outcomes having the required number of successes and failures

Probability that there are x successes and (n-x) failures

  - The mean, $E(x) = p + p + \ldots + p = n*p$
  - The variance, $V(X) = pq + pq + \ldots + pq = n*pq$

Geometric & Negative Binomial Distribution

- Geometric distribution
  - The number of Bernoulli trials, *X*, to achieve the 1st success:

$$p(x) = \begin{cases} q^{x-1} p, & x = 0,1,2,\ldots,n \\ 0, & \text{otherwise} \end{cases}$$

  - $E(x) = 1/p$, and $V(X) = q/p^2$

- Negative binomial distribution
  - The number of Bernoulli trials, X, until the kth success
  - If Y is a negative binomial distribution with parameters p and k, then:

$$p(y) = \begin{cases} \binom{y-1}{k-1} q^{y-k} p^k, & y = k, k+1, k+2,\ldots \\ 0, & \text{otherwise} \end{cases}$$

  - $E(Y) = k/p$, and $V(X) = kq/p^2$

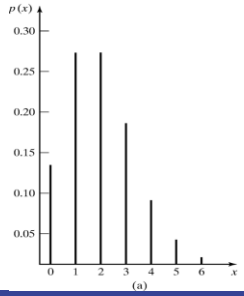
Poisson Distribution

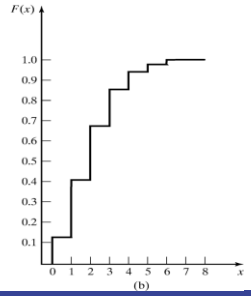
- Poisson distribution describes many random processes quite well and is mathematically quite simple.
  - where $\alpha > 0$, ***pdf*** and ***cdf*** are:

$$p(x) = \begin{cases} \dfrac{e^{-\alpha}\alpha^{x}}{x!}, & x = 0,1,... \\ 0, & \text{otherwise} \end{cases} \qquad F(x) = \sum_{i=0}^{x} \frac{e^{-\alpha}\alpha^{i}}{i!}$$

  - $E(X) = \alpha = V(X)$

Poisson random variable x = the ***number of occurrences*** of a specified event in a ***given unit of time or space***.





Poisson Distribution

- Example: A computer repair person is "beeped" each time there is a call for service. The number of beeps per hour ~ Poisson(a = 2 per hour).
  - The probability of three beeps in the next hour:

    $p(3) = e^{-2}2^3/3! = 0.18$

    also, $p(3) = F(3) - F(2) = 0.857-0.677=0.18$
  - The probability of two or more beeps in a 1-hour period:

    $p(2 \text{ or more}) = 1 - p(0) - p(1)$

    $= 1 - F(1)$

    $= 0.594$

CONTINUOUS DISTRIBUTIONS

Continuous Distributions

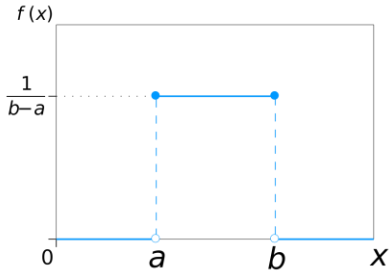
- Continuous random variables can be used to describe random phenomena in which the variable can take on any value in some interval.
- In this section of the lecture, the distributions studied are:
  - Uniform
  - Exponential
  - Normal
  - Weibull
  - Lognormal

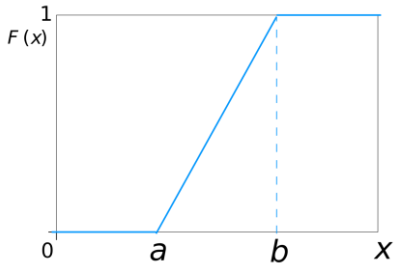
Uniform Distribution

- A random variable X is uniformly distributed on the interval (a,b), $U(a,b)$, if its ***pdf*** and ***cdf*** are:

$$f(x)=\begin{cases}\dfrac{1}{b-a}, & a\le x\le b\\ 0, & \text{otherwise}\end{cases}$$

$$F(x)=\begin{cases}0, & x\prec a\\ \dfrac{x-a}{b-a}, & a\le x\prec b\\ 1, & x\ge b\end{cases}$$





Uniform Distribution

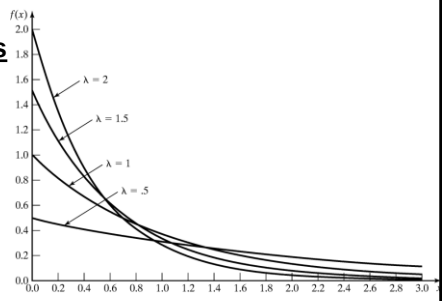
- Properties
  - $P(x_1 < X < x_2)$ is proportional to the length of the interval $[F(x_2) - F(x_1) = (x_2-x_1)/(b-a)]$
  - $E(X) = (a+b)/2$ $\quad V(X) = (b-a)^2/12$
- U(0,1) provides the means to generate random numbers, from which random variates can be generated.

Exponential Distribution

- A random variable X is exponentially distributed with parameter $\lambda > 0$ if its ***pdf*** and ***cdf*** are:

$$f(x) = \begin{cases} \lambda e^{-\lambda x}, & x \geq 0 \\ 0, & \text{elsewhere} \end{cases} \qquad F(x) = \begin{cases} 0, & x < 0 \\ \int_0^x \lambda e^{-\lambda t} dt = 1 - e^{-\lambda x}, & x \geq 0 \end{cases}$$

  - $E(X) = 1/\lambda$ $\qquad$ $V(X) = 1/\lambda^2$
  - **Used <u>to model interarrival times</u> when arrivals are completely random**, *and* **<u>to model service times</u> that are highly variable**
  - For several different exponential ***pdf***s (see figure), the value of intercept on the vertical axis is λ, and all pdf's eventually intersect.



Exponential Distribution

- Memoryless property
  - For all s and t greater or equal to 0:

$$P(X > s+t \mid X > s) = P(X > t)$$

  - Example: A lamp ~ exp(λ = 1/3 per hour), hence, on average, 1 failure per 3 hours.
    - The probability that the lamp lasts longer than its mean life is:

$$P(X > 3) = 1-(1-e^{-3/3}) = e^{-1} = 0.368$$

    - The probability that the lamp lasts between 2 to 3 hours is:

$$P(2 <= X <= 3) = F(3) - F(2) = 0.145$$

    - The probability that it lasts for another hour given it is operating for 2.5 hours:

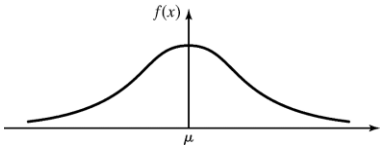
$$P(X > 3.5 \mid X > 2.5) = P(X > 1) = e^{-1/3} = 0.717$$

Normal Distribution

- A random variable X is normally distributed has the pdf:

$$f(x)=\frac{1}{\sigma\sqrt{2\pi}}\exp\left[-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2\right],\ -\infty \prec x \prec \infty$$

  - Mean: $-\infty \prec \mu \prec \infty$
  - Variance: $\sigma^2 \succ 0$
  - Denoted as $X \sim N(\mu,\sigma^2)$



- Special properties:
  - $\lim_{x\to-\infty} f(x)=0$, and $\lim_{x\to\infty} f(x)=0$
  - $f(\mu-x)=f(\mu+x)$; the pdf is symmetric about μ.
  - The maximum value of the ***pdf*** occurs at $x = \mu$; the mean and mode are equal.

Normal Distribution

- Evaluating the distribution:
  - Use numerical methods (no closed form)
  - Independent of μ and σ, using the standard normal distribution:

$$Z \sim N(0,1)$$

  - Transformation of variables: let Z = (X - μ) / σ,

$$F(x)=P(X\le x)=P\left(Z\le\frac{x-\mu}{\sigma}\right)$$

$$=\int_{-\infty}^{(x-\mu)/\sigma}\frac{1}{\sqrt{2\pi}}e^{-z^2/2}dz$$

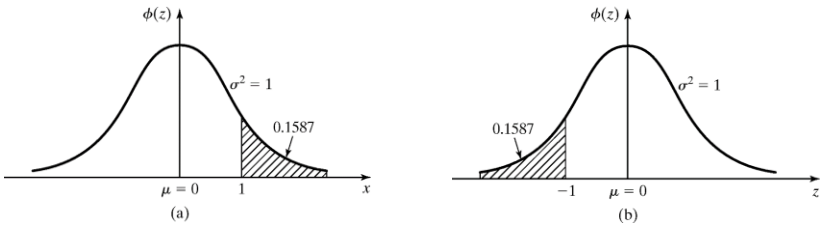
$$=\int_{-\infty}^{(x-\mu)/\sigma}\phi(z)dz=\Phi(\tfrac{x-\mu}{\sigma}) \quad \text{, where } \Phi(z)=\int_{-\infty}^{z}\frac{1}{\sqrt{2\pi}}e^{-t^2/2}dt$$

Normal Distribution

- Example: The time required to load an oceangoing vessel, *X*, is distributed as *N(12,4)*
  - The probability that the vessel is loaded in less than 10 hours:

$$F(x<10)=\Phi\left(z<\frac{10-12}{2}\right)=\Phi(z<-1)=0.1587$$

  - Using the symmetry property, $\Phi(1)$ is the complement of Φ (-1)



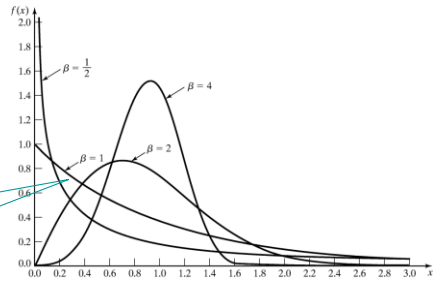
Weibull Distribution

- A random variable *X* has a Weibull distribution if its ***pdf*** has the form:

$$f(x)=\begin{cases}\frac{\beta}{\alpha}\left(\frac{x-\nu}{\alpha}\right)^{\beta-1}\exp\left[-\left(\frac{x-\nu}{\alpha}\right)^{\beta}\right], & x\geq\nu\\ 0, & \text{otherwise}\end{cases}$$

- 3 parameters:
  - Location parameter: υ, $(-\infty \prec \nu \prec \infty)$
  - Scale parameter: β, $(\beta > 0)$
  - Shape parameter. α, $(\alpha > 0)$
- Example: $\upsilon = 0$ and $\alpha = 1$:

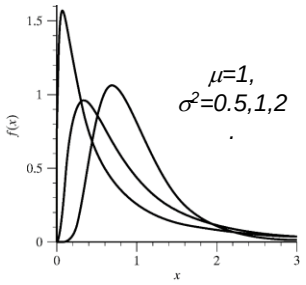
When $\beta = 1$, $X \sim exp(\lambda = 1/\alpha)$



Lognormal Distribution

- A random variable X has a lognormal distribution if its pdf has the form:

$$f(x) = \begin{cases} \frac{1}{\sqrt{2\pi}\sigma x} \exp\left[-\frac{(\ln x - \mu)^2}{2\sigma^2}\right], & x > 0 \\ 0, & \text{otherwise} \end{cases}$$



  - Mean $E(X) = e^{\mu+\sigma^2/2}$
  - Variance $V(X) = e^{2\mu+\sigma^2/2}(e^{\sigma^2} - 1)$

- Relationship with normal distribution
  - When $Y \sim N(\mu, \sigma^2)$, then $X = e^Y \sim$ lognormal(μ, σ^2)
  - Parameters μ and σ^2 are not the mean and variance of the lognormal

POISSON DISTRIBUTION

Poisson Process and Distribution

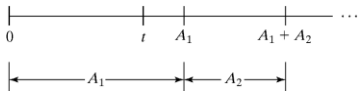
- Definition: *N(t)* is a counting function that represents the number of events occurred in *[0,t]*.
- A counting process {*N(t), t>=0*} is a Poisson process with mean rate λ if:
  - Arrivals occur one at a time
  - {*N(t), t>=0*} has stationary increments
  - {*N(t), t>=0*} has independent increments
- Properties

$$P[N(t)=n]=\frac{e^{-\lambda t}(\lambda t)^n}{n!}, \qquad \text{for } t \geq 0 \text{ and } n = 0,1,2,...$$

  - Mean and variance are equal : $E[N(t)] = V[N(t)] = \lambda t$
  - Stationary increment: The number of arrivals in time *s* to *t* is also Poisson-distributed with mean $\lambda(t-s)$

Interarrival Times

- Consider the interarrival times of a Possion process (A_1, A_2, ...), where A_i is the elapsed time between arrival *i* and arrival *i+1*



  - The 1st arrival occurs after time t iff there are no arrivals in the interval [0,t], hence:

$$P\{A_1 > t\} = P\{N(t) = 0\} = e^{-\lambda t}$$
$$P\{A_1 <= t\} = 1 - e^{-\lambda t} \qquad \text{[cdf of exp(λ)]}$$

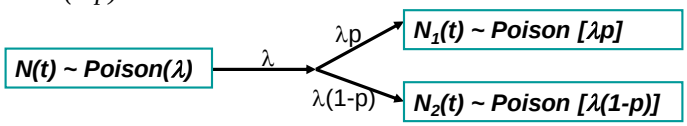
  - Interarrival times, A_1, A_2, ..., are exponentially distributed and independent with mean $1/\lambda$

Arrival counts ~ *Poison(λ)* ⟺ Interarrival time ~ *Exp(1/λ)*

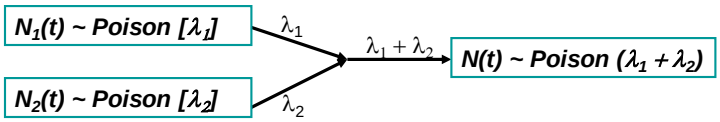
Stationary & Independent — Memoryless

Splitting and Pooling

- Splitting:
  - Suppose each event of a Poisson process can be classified as Type I, with probability *p* and Type II, with probability *1-p*.
  - $N(t) = N_1(t) + N_2(t)$, where $N_1(t)$ and $N_2(t)$ are both Poisson processes with rates λp and $\lambda(1-p)$



- Pooling:
  - Suppose two Poisson processes are pooled together
  - $N_1(t) + N_2(t) = N(t)$, where $N(t)$ is a Poisson processes with rates $\lambda_1 + \lambda_2$



Nonstationary Poisson Process (NSPP)

- Poisson Process without the stationary increments, characterized by $\lambda(t)$, the arrival rate at time *t*.
- The expected number of arrivals by time *t*, $\Lambda(t)$:

$$\Lambda(t) = \int_0^t \lambda(s)ds$$

- Relating stationary Poisson process *n(t) with rate* $\lambda=1$ and NSPP *N(t)* with rate $\lambda(t)$:
  - Let arrival times of a stationary process with rate $\lambda = 1$ be $t_1, t_2, \ldots$, and arrival times of a NSPP with rate $\lambda(t)$ be $T_1, T_2, \ldots$, we know:

$$t_i = \Lambda(T_i)$$
$$T_i = \Lambda^{-1}(t_i)$$

It means that PP can be transformed to NSPP with arrival rate $\lambda=1$, *and NSPP can be transformed to PP with arrival rate* $\lambda(t)$.

Empirical Distributions

- A distribution whose parameters are the observed values in a sample of data.
  - May be used when it is impossible or unnecessary to establish that a random variable has any particular parametric distribution.
  - Advantage: no assumption beyond the observed values in the sample.
  - Disadvantage: sample might not cover the entire range of possible values.

Summary

- The world that the simulation analyst sees is probabilistic, not deterministic.
- In this lecture:
  - Reviewed several important probability distributions.
  - Showed applications of the probability distributions in a simulation context.
- Important task in simulation modeling is the collection and analysis of input data, e.g., hypothesize a distributional form for the input data. Modelers (students) should know:
  - Difference between discrete, continuous, and empirical distributions.
  - Poisson process and its properties.